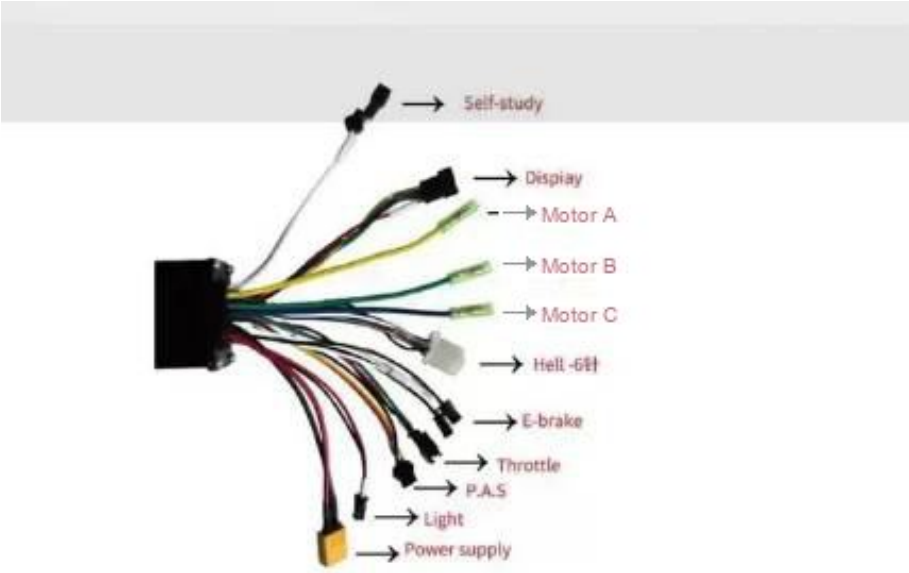
E-Bike Brushless Controller

Full Wiring & Setup Guide

Reference Guide for Electric Vehicle Systems



Controller cable layout: Power supply, Motor phase, Hall sensor, Display, Throttle, P.A.S, E-brake, Light, Self-study

This manual provides step-by-step instructions for the correct wiring and initial configuration of your brushless motor controller. Read it completely before you connect anything.

Document version 2 - supersedes all earlier revisions

Rated voltage and current: see the label on your controller. Never connect a battery that exceeds the rating on the label.

Keep this guide with your bike for future reference.

Quick start

1. Battery (yellow XT60) - check the voltage and the polarity first
2. Motor phase wires (yellow A / green B / blue C) - match the colours
3. Display (5-pin) - Protocol 2, align the pins
4. Self-study white wires - connect once to calibrate, then unplug
5. Motor Hall sensor (6-pin white) - press fully home until it latches
6. Throttle (3-pin) - red +5V, green signal, black ground
7. PAS sensor (3-pin) - leave disconnected if you have no pedal assist

Then work through the pre-ride checklist in section 4 before riding.

1. Safety Warnings

Read this section before you connect anything. Most controller failures, and all reported fire incidents, are avoidable.

- **Disconnect the battery first.** Never connect or disconnect any connector while the battery is connected. Live-plugging can damage the controller and the harness.
- **Never exceed the rated voltage marked on the controller label.** A battery of higher voltage than the controller's rating will overheat the controller and the wiring, and it can cause a fire. Check the battery output voltage before you connect it.
- **Never reverse the battery polarity.** Red is positive (+); black is negative (-). Reversed polarity destroys the controller immediately.
- **Match the controller to your motor and battery.** Use the controller only within the rated voltage and power marked on its label.
- **Do not modify the harness.** Do not splice, extend or replace wires with thinner wire or with lower-rated connectors. Use only the harness supplied with the controller.
- **Route and secure every cable.** Keep cables clear of sharp edges, pinch points and rotating parts, and secure them with cable ties so that they cannot chafe.
- **Do not open the housing.** There are no user-serviceable parts inside. Opening the housing voids the warranty.
- **Keep the controller away from water.** Mount it where it cannot be soaked or submerged. Water ingress causes short circuits.

STOP USING THE CONTROLLER IMMEDIATELY AND DISCONNECT THE BATTERY IF YOU NOTICE ANY OF THE FOLLOWING:

- a burnt smell, smoke, or soot
- wire insulation that is discoloured, hardened, brittle or melted
- a housing or connector that becomes unusually hot in normal use
- an error code that comes back after you have disconnected and reconnected the wiring
- sparks, a clicking relay, or the motor locking up
- any wire or connector that has come loose at an internal joint

If a wire or connector shows heat damage, replace the harness. Do not tape over damaged insulation and continue to ride.

What is new in version 2

- Added this safety section, including the rated-voltage warning.
- Corrected the illustration for each installation step.
- Added a pre-ride checklist (section 4).
- Expanded the error-code table and added detailed troubleshooting steps (sections 5 and 6).
- Added rated-limits, package-contents and support information (sections 7 to 10).

2. Wiring Quick Reference

Match every connector to the table below before you power the system. Wire colours refer to the harness supplied with this controller. If your harness differs, contact support before connecting anything.

Function	Connector / wire description	Notes
Power supply (battery)	Yellow XT60; thick red (+) and thick black (-)	Check polarity before connecting. Never reverse it.
Motor phase	Thick yellow (A), green (B), blue (C); round bullet terminals	Match the colours to your motor phase wires, tighten the terminals, and fit insulation sleeves.
Display / instrument	5-pin SM black (red, blue, black, green, yellow)	Protocol 2 communication. Align the pins before pressing.
Hall sensor (motor)	6-pin white rectangular; red = +5V, black = ground, yellow / green / blue = Hall A / B / C	Must be pressed fully home until it latches.
Throttle	3-pin SM black; red = +5V, green = signal, black = ground	Confirm the wire order matches your throttle pinout.
PAS sensor	3-pin SM black; red = +5V, yellow = signal, black = ground	Leave disconnected if your bike has no pedal assist.
Self-study	Two thin white wires, single-pin connectors	Connect only to run auto-calibration, then unplug permanently.
E-brake (optional)	Brake cut-off input	Leave disconnected if your bike has no brake sensor.
Light (optional)	Light output, where fitted	Optional equipment. Leave disconnected if not used.

Identifying the connectors

- The controller housing carries a printed label for each connector group: Power supply, Motor phase, Hall sensor, Display, Throttle, P.A.S, E-brake and Light.
- Some labels use the symbols Hall+5V for the sensor supply (red wire) and Hall A / Hall B / Hall C for the three position signals.
- If a label is unreadable or a connector is missing, stop and contact support. Do not guess a connection.

3. Installation Steps

Complete the steps in order. Power the system off before making or breaking any connection.

Step 1: Power Supply Connection

Connect the battery to the controller.

Wire details: Thick red (+) and thick black (-) wires with a yellow XT60 connector.

- Confirm the battery output voltage does not exceed the rating on the controller label.
- Push the XT60 connectors fully together until they latch. They are keyed and fit only one way - never force them.
- Never connect or disconnect the XT60 while the battery is switched on.
- Check: the controller powers up when the power-lock signal is active.



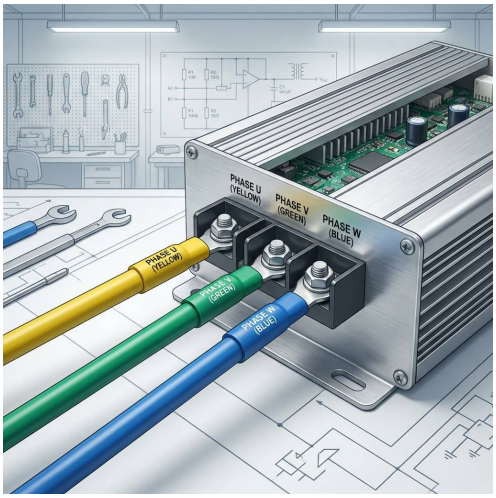
Step 1 - battery and XT60 main power connector

Step 2: Motor Phase Line Connection

Connect the high-power phase lines to the motor.

Wire details: Thick yellow (A), green (B) and blue (C) wires with round bullet terminals.

- Match the colours to your motor phase wires and tighten every terminal firmly.
- Fit an insulation sleeve over every terminal.
- Never leave a phase terminal loose or exposed: a loose phase connection arcs, overheats and can melt the wiring.
- If the motor runs backwards after the self-study step, swap any two phase wires and repeat the self-study step.



Step 2 - motor phase terminals (U / V / W)

Step 3: Instrument / Display Connection

Plug in the display to activate the system.

Wire details: 5-pin SM black connector containing red, blue, black, green and yellow wires. This uses Protocol 2 communication.

- Power the system off. Align the pins and press the connector fully home until it latches.
- Do not pull on the cable to disconnect it; release the latch first.
- An E10 or E11 code means this connector or its wiring is not making contact.



Step 3 - display connector and instrument

Step 4: Self-Learning Configuration

Auto-calibrate the motor phase and direction.

Wire details: Two thin white wires with single-pin connectors.

- With the system powered, connect the two white wires together. The motor will begin to rotate slowly.
- Let it rotate until the rotation is smooth and in the correct direction. If the direction is wrong, swap any two motor phase wires and repeat.
- Disconnect the white wires and leave them disconnected permanently. Insulate the ends.
- Never leave the self-study wires connected during normal riding.



Step 4 - self-study white wires

Step 5: Motor Hall Sensor Connection

Provide rotor position feedback for smooth motor operation.

Wire details: 6-pin white rectangular connector: red = +5V supply, black = ground, yellow / green / blue = Hall A / B / C.

- Line up the keyway and press the connector fully home until it clicks. A partly seated Hall connector is the most common cause of a stuttering motor and of the E07 and E10 error codes.
- Do not pull on the cable to disconnect it.
- Inspect the whole length of the cable for a pinch, a rub or a crushed section, especially where it passes the axle.



Step 5 - 6-pin Hall sensor connector

Step 6: Throttle Connection

Connect the speed control handle.

Wire details: 3-pin SM black connector: red (+5V), green (signal), black (ground).

- Confirm the wire order matches your throttle pinout before connecting.
- The throttle must return freely to the closed position. If it does not, do not ride the bike.
- Test the throttle with the drive wheel off the ground before your first ride.



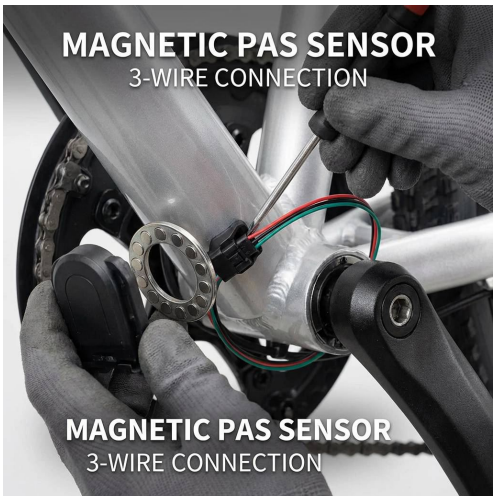
Step 6 - throttle grip and 3-pin connector

Step 7: Pedal Assist System (PAS) Connection

Enable the pedalling assist function.

Wire details: 3-pin SM black connector: red (+5V), yellow (signal), black (ground).

- Mount the sensor on the crank with the magnet disc parallel to and close to the sensor, with a small air gap.
- Plug the connector in and check that the assist cuts in and out smoothly.
- If your bike has no pedal assist, leave this connector disconnected.



Step 7 - magnetic PAS sensor

4. Pre-Ride Checklist

Carry out every check before the first ride and after any service to the electrical system.

[ ] Battery voltage does not exceed the rating on the controller label
[ ] Battery polarity correct: red to red, black to black
[ ] Motor phase terminals tight and covered by insulation sleeves
[ ] Hall sensor connector fully seated and latched
[ ] Display connector fully seated and latched
[ ] Throttle and PAS connectors fully seated
[ ] Self-study white wires disconnected and insulated
[ ] Every cable secured and clear of sharp edges, pinch points and rotating parts
[ ] Controller mounted so that it can dissipate heat and cannot be soaked
[ ] With the drive wheel off the ground: power on, confirm no error code is shown, and check the throttle responds smoothly
[ ] Run the bike at low load for a few minutes, then feel the controller and the connectors for abnormal heat before riding normally

5. Troubleshooting & Error Codes

If the display shows a code, find it in the table below and then follow the detailed steps in section 6.

Code	Meaning	What to do
E02	Brake fault / brake cut-off active	Check that the brake lever has returned fully to its original position. Clear any obstruction, or replace the brake sensor.
E06	Low power / voltage fault	Check the battery charge level and confirm the controller receives the correct input voltage.
E07	Motor / phase problem	Check the motor wiring and the Hall sensor connector. Confirm the motor is not physically stuck. If the code persists, test with a different controller or motor.
E08	Throttle (handle) problem	Check that the throttle returns to the closed position and check its signal.
E09	Controller problem	Potential MOSFET damage inside the controller. The controller usually needs to be replaced.
E10	Data receiver failure	Communication loss (Rx) between the display and the controller. Check the display connector and wiring.
E11	Data sender failure	Communication loss (Tx) between the display and the controller. Check the display connector and wiring.

- **A code that is not listed above** is not documented in this manual. Record the code, disconnect the battery, stop riding, and contact support. Do not continue to ride with an unresolved code.
- **A code that comes back after you disconnect and reconnect the wiring** does not mean the reset failed. It means the fault is inside a cable, a connector crimp, the motor, or the controller. Reconnecting the plugs will not clear it - the fault has to be found and repaired.

E02: Brake fault. Check whether the brake handle has returned to its original position. Clear any obstruction, or replace the brake sensor if necessary.

E06: Voltage fault. Verify the battery charge level and ensure the controller receives the correct input voltage.

E07: Motor / phase fault. Check the motor wiring and the Hall sensor connections. Ensure the motor is not physically stuck. If the code persists, test with a different controller or motor.

E09: Controller MOS fault. Internal component failure. The controller typically needs to be replaced.

E10 / E11: Communication fault. Check the instrument plug and the wiring between the display and the controller. Ensure the electric door lock signal is active.



The display reports a diagnostic code - in this example, E07 (motor / phase problem)

6. Step-by-Step Troubleshooting

6.1 The controller does not power on

- Check the battery: measure the output voltage. It must be within the range marked on the controller label and above the low-voltage cut-off.
- Check the power-lock (ignition) signal. The controller cannot power on if the power-lock line is not connected.
- Check the main fuse or the battery management system: a tripped BMS gives no output.
- Disconnect the battery, then re-seat every connector and try again.
- If the controller still does not power on and the battery voltage is correct, contact support with your order number and the code shown on the display.

6.2 The motor stutters, runs roughly, or will not accelerate

- Disconnect the battery first, then check the 6-pin Hall sensor connector: it must be pressed fully home until it latches. A partly seated connector can make contact intermittently as the motor vibrates.
- Check the red wire in the Hall connector: it supplies +5V to the motor Hall sensors. If it is not seated or not crimped correctly, the motor receives no position signal and will stutter or fail to run.
- Check the motor phase wires: every terminal must be tight and insulated, and no phase wire may be pinched or rubbed through.
- Check the throttle connector: red (+5V), green (signal), black (ground).
- Disconnect the self-study white wires if they are still connected.
- If the display shows E07 or E10, go to section 6.3.

6.3 Hall signal test

Carry out this test only if you can use a multimeter safely. If you cannot, contact support instead.

- Disconnect the battery. Then disconnect the 6-pin Hall connector at the controller.
- Reconnect the battery and power the controller on.
- With the multimeter on DC volts, measure between the +5V pin (red) and the ground pin (black) of the Hall connector. You should read close to 5V. If you read 0V, the sensor supply is not reaching the connector - check the red wire and the connector crimp.
- Measure each of the three Hall signal pins (yellow, green, blue) against ground while you slowly rotate the wheel by hand. Each signal must switch between about 0V and about 5V as the wheel turns, in three different patterns.
- If any signal stays at 0V or stays at 5V while the wheel turns, that Hall circuit is faulty - in the motor, in the cable, or in the connector.
- Disconnect the battery before you reconnect the Hall connector.
- If the 5V supply is present and all three signals switch correctly but the error code returns, the fault is inside the controller - contact support.

6.4 An error code returns after you reconnect all the wiring

The cable or the connector is damaged internally, or the fault is inside the motor or the controller. Reconnecting the plugs will not fix it.

- Disconnect the battery.
- Inspect the full length of the motor cable for a pinch, a rub or a crushed section, especially where it passes the axle and the frame.
- Inspect the back of every connector for a wire that has pulled out of its crimp. Gently tug each wire - it must not move inside the connector.
- If the cable and connectors look intact, contact support with the code and your order number. Do not keep riding while an error code is shown.

6.5 The controller or the wiring gets hot, smells burnt, or the insulation is melted

STOP IMMEDIATELY.

- Disconnect the battery and do not use the bike.
- Do not attempt to repair the controller or the harness yourself.
- Photograph the affected part.
- Contact support with your order number, the photographs and the code shown on the display. If the unit is within warranty, we will replace it.

Do not continue to use a controller that has overheated, and do not fit a controller whose wiring is discoloured or melted. Continuing to use it can lead to a fire.

6.6 When to contact support

Contact us with your order number, the code shown on the display, and photographs if:

- any error code returns after you have reconnected the wiring
- the controller or the wiring becomes hot, smells burnt, or looks melted
- the controller does not power on and the battery voltage is correct
- the motor does not turn after the checks in section 6.3
- you are not able to carry out any of the checks above safely

7. Rated Limits and Identification

Item	Specification	Note
Maximum input voltage	The value printed on the controller label	Never exceed it. A higher-voltage battery overheats the controller and the wiring and can cause a fire.
Rated current / power	The value printed on the controller label	Do not operate the controller above its rated current or power.
Motor type	Three-phase brushless motor with Hall sensors	This controller requires a motor with a Hall sensor cable.
Control protocol	Protocol 2	The display must use Protocol 2 communication.
Mounting	On a surface that can dissipate heat	Keep the controller away from water and do not cover it or block airflow around it.

If the label on your controller is missing or unreadable, contact support and confirm the correct voltage before you connect a battery. Connecting a battery above the rated voltage will overheat the controller and the wiring and can cause a fire.

8. Package Contents

Your kit may not contain every item listed below. Check the parts you received against your order.

- Controller unit with mounting hardware
- Main power harness with XT60 connector
- Motor phase cable with bullet terminals
- 6-pin motor Hall sensor cable
- 5-pin display / instrument cable
- 3-pin throttle
- 3-pin PAS sensor (where supplied)
- Self-study white wires
- This guide

- If any part is missing or damaged on arrival, contact support before installation. Do not substitute a different harness or connector.
- Keep the packaging and this guide until installation is complete.

9. Maintenance and Storage

- Check the connectors and the condition of the cables before every ride, and re-seat any connector that is not fully latched.
- Keep the controller clean and dry. Do not pressure-wash it, and do not allow water to pool around it.
- Do not cover the controller or block the airflow around its housing.
- Inspect the harness for chafing where it passes the frame and the axle, and re-secure it with cable ties if it has moved.
- Store the bike and the battery as instructed in the battery manufacturer's own manual.

10. Warranty and Support

- This product is covered by the warranty stated on the product listing.
- To obtain support, message the seller with your order number, the code shown on the display, and photographs of the affected part.
- Stop using the product and contact us before attempting any repair.
- Retain this guide and your order details for any warranty claim.

11. Important Notice

This guide covers the controller model supplied with your order. Connector types, wire colours and specifications can differ between models. Always follow the rated limits printed on your own controller and, if anything in this guide does not match the parts you received, stop and contact support before connecting a battery.

Do not modify the controller or the harness. Any modification, and any use outside the rated voltage, current or power, voids the warranty and can create a fire hazard.

Read the entire guide before you begin. If any step is unclear, stop and contact support before continuing.